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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Metamind
Selected Domain	Domain 1 - The Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 4 (Quality Education) & SDG 8 (Decent Work & Economic Growth)
Project / Solution Name	SkillShift MY
Name (Leader)	Sallam Maram Niyazi Abdulmawla
Email & Phone (Leader)	nazmaram948@gmail.com/ +601139774316
University Affiliation(s)	UniKL MIIT/ XMUM
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Artificial intelligence is transforming labor markets faster than traditional training frameworks can adapt. In Malaysia, 56% of professionals use AI at work, yet only 26% have formal training (Hays Asia Salary Guide 2025), and the Ministry of Science, Technology and Innovation has said more than 600,000 workers will need reskilling within three to five years. This affects working adults at risk of job displacement, fresh graduates and career switchers, HR and learning-and-development managers, and TVET and university career centres.

The core problem is that career guidance relies on static, disconnected lists that become obsolete as AI evolves. Individuals cannot tell which skills are rising or declining, employers struggle to decide which roles to train for, and few tools link skill development to the UN Sustainable Development Goals. Without dynamic, data-driven foresight, reskilling stays reactive, worsening talent mismatches and economic vulnerability. This undermines SDG 4 (Quality Education), specifically relevant skills for decent jobs (Target 4.4) and lifelong technical and vocational learning (Target 4.3), and SDG 8 (Decent Work and Economic Growth), specifically productivity through innovation (Target 8.2) and full and productive employment (Target 8.5). Malaysia needs a continuously updated, transparent tool that helps people navigate AI-driven workforce transitions with clarity, while leaving the decision to them.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SkillShift MY is an AI-assisted career-transition web app that takes a person from "who I am today" to "where I can be in the AI era". Instead of a static job list or a chatbot, it combines public data (ESCO, O*NET and the Anthropic Economic Index) in a Python analytics engine, and every step shows its source and date.

It works in four steps. First, the user uploads a synthetic CV, pastes text or picks a demo profile; a small pre-trained embedding model (sentence-transformers all-MiniLM-L6-v2) matches the text to ESCO skills, and the user confirms each skill. Second, a Future Skills Radar shows rising and declining skills computed from Anthropic Economic Index snapshots. Third, 42 roles are ranked with an explainable score (skill coverage, role outlook and transferability) and a skills gap, and a 1,000-run Monte Carlo simulation, varying skill momentum by up to 20%, estimates how often each role stays in the user's top 5. Fourth, every role is linked to the SDGs, an SDG slider shows how the ranking changes, and a learning pathway lists catalogue courses for the skill gaps, with hours shown only when verified.

Loading a newer data snapshot visibly changes the rankings and explains why. To avoid invented content, every skill, role, course and SDG ID is checked against the local database and unknown IDs are blocked, and no external generative-AI API is used. All forecasts are labelled Estimate, and the person makes the final decision. From the SDC Phase 1 workshops we apply database architecture (SQLite), API design (FastAPI) and cloud hosting (Replit) .

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

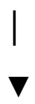
Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Technologies, Frameworks, APIs & Databases:

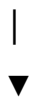
- **Frontend:** React, Vite and Recharts. Skills Radar, role cards with score breakdowns, SDG explorer and slider, and learning pathway.
- **Backend:** FastAPI (Python) serving /api/* endpoints, CV text extraction (pdfplumber, 20,000-character cap), strict schemas and the database ID guardrail.
- **Engine:** pandas and NumPy; pre-trained embedding model sentence-transformers all-MiniLM-L6-v2 (CPU) for skill matching; skill momentum, role scoring and the 1,000-run Monte Carlo robustness estimate. -
- **Data:** SQLite database built from ESCO v1.2, O*NET 31.0, Anthropic Economic Index snapshots (Nov 2025, May 2026), a team-curated course catalogue (60 courses) and SDG mapping (42 occupations), and synthetic personas.
- **AI and guardrails:** no external generative-AI API at runtime. An optional LLM interface exists in the code and is disabled.-
- **Cloud and tools:** Replit for hosting, GitHub for version control; Claude Code, Claude and the Replit Agent for AI-assisted development (declared in the Technical & Tools Declaration).

System Architecture Block Diagram:

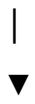
[User: synthetic CV / pasted text / demo profile]



[React / Vite frontend: Skills Radar, Role Cards, SDG slider, Learning Pathway]



[FastAPI backend on Replit: CV text extraction, ID guardrail]



[Python engine: embedding skill match, momentum, role scoring, 1,000-run Monte Carlo estimate]



[SQLite: ESCO, O*NET, AEI, course and SDG tables]

Sponsor-Provided Resources Usage:

- **Official SDC Sponsor Resources Used: Yes.** The project utilizes **Replit Core** for hosting, cloud deployment, and live prototype availability as required by hackathon guidelines.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We applied design thinking to keep the solution viable and user-friendly. Empathise: the Domain 1 brief names four users (working adults at risk of displacement, fresh graduates and career switchers, HR and L&D managers, and TVET and university career centres), and we focused on working adults. Define: they do not know which skills to learn, and static lists go out of date. Ideate and prototype: instead of a chatbot, we built a transparent journey in which the person confirms their skills, sees the

source and date of every signal, and decides for themselves. Test: we ran synthetic personas (an accounts clerk, a customer-service officer, a data-entry clerk and a fresh graduate) through the full flow, fixed the data and wording problems we found, and added automated tests, including a 50-run guardrail test. For deployment, the system is a React frontend, a FastAPI backend and a SQLite database in one Replit app, rebuilt by a documented build step, with a public prototype link ready for Phase 2 work such as verified course data and a measured SDG score.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 4:** Quality Education. Target 4.4: Substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship. Target 4.3: Ensure equal access for all women and men to affordable and quality technical, vocational and tertiary education. **SDG 8:** Decent Work and Economic Growth. Target 8.2: Achieve higher levels of economic productivity through diversification, technological upgrading and innovation. Target 8.5: Achieve full and productive employment and decent work for all women and men, including young people and persons with disabilities.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**Training gap:** in Malaysia, 56% of professionals use AI tools at work, yet only 26% have received formal training (Hays Asia Salary Guide 2025). **Workforce vulnerability:** more than 600,000 Malaysian workers will need reskilling within three to five years (Ministry of Science, Technology and Innovation, as cited in the Domain 1 brief). **Static guidance:** the Domain 1 brief notes that learning pathways are usually static lists that are out of date by the time they are published.
Core Measurable KPI *The specific unit you will track*	**1. Skill-match gain per verified training hour:** the gain in a role's skill-coverage match score per verified course hour (computed by the engine as marginal score per hour, shown only where the catalogue covers the person's skill gaps).

Metric Category	Your Response
e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.	**2. Role robustness:** the share of 1,000 simulated futures in which a role stays in the user's top 5. **3. Guardrail compliance:** the share of API responses whose skill, role, course and SDG IDs exist in the database (50-run guardrail test, target 100%).
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	**Targets to validate with test users in Phase 2:** - At least 30% higher skill-match score for users who follow a SkillShift pathway, compared with their starting profile .- Up to 25% less study time by prioritising courses with the highest gain per verified hour.- 100% of returned IDs verified against the database. We will also expand the course catalogue and verify course costs and HRD Corp claimability.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✔️ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

✔️ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
ESCO v1.2	Dataset	CC BY 4.0
O*NET 31.0	Dataset	CC BY 4.0
Anthropic Economic Index (AEI)	Dataset	Open Access / Public
sentence-transformers all-MiniLM-L6-v2	Dataset	Open Government Data (Malaysia)
sentence-transformers all-MiniLM-L6-v2	Pre-trained model	Apache-2.0
React, Vite, Recharts	Code	MIT
FastAPI, Uvicorn, Pydantic	Code	MIT; BSD-3-Clause; MIT
pandas, NumPy, PyArrow	Code	BSD-3-Clause; BSD-3-Clause Apache-2.0
sentence-transformers, PyTorch (CPU)	Code	Apache-2.0; BSD-3-Clause
SQLite	Code	Public Domain
pdfplumber	Code	MIT
requests, openpyxl, xlrd, PyYAML, python-dotenv, pytest, httpx, ESLint	Code	MIT, BSD-3-Clause or Apache-2.0
Lexend font (Google Fonts)	Font	SIL Open Font License 1.1

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✔️ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24.10.2026